RoboCup Humanoid Soccer League (HSL) Rule Book

RoboCup Humanoid Soccer League Technical Committee

(DRAFT 2026 Working Rules Document, as of 2026-01-31)

Questions or comments on these rules should be submitted via the following channels:

- `https://github.com/RoboCup-HumanoidSoccerLeague/HSL-Rules/issues`;
- Discord server: `https://discord.gg/67upw6Nn`, channel `#rule-book`;
- Email to `tba@tba.com`.

Contents

1 Purpose and Scope of the RoboCup Humanoid Soccer League

This document defines the purpose, scope, and rules of the RoboCup Humanoid Soccer League for the international RoboCup competition, at the forefront of research, innovation, and education in humanoid soccer across all forms.

1.1 Vision and Mission Statement

The RoboCup Humanoid Soccer League (HSL) will drive innovative research that advances the software and hardware of autonomous robots with a particular emphasis on robots deployed in real-time, dynamic, partially observable, and multi-agent environments. The HSL is well suited to advance *research*, *development*, and *education* in:

- Multi-robot systems (5+ robots) requiring decentralized coordination with limited communication over noisy channels.
- Robots that approach or approximate Human-like capabilities.
- A league that promotion research,
- Localization and state-estimation.
- Dynamic humanoid motor control.
- Real-time and on-board robot perception.
- Software engineering for autonomous robots.
- Hardware engineering for autonomous robots.
- Across topics, robot learning in all its forms.

In addition, the HSL aim to:

- Grow the community of humanoid soccer within RoboCup.
- Further education in robotics and is designed such that both teams with a primary focus in research and teams with a primary focus in education are able to participate.
- Encourage active sharing of software and hardware designs for league-wide collaboration.
- Measure the capabilities of the league against the 2050 vision of RoboCup.
- Drive the vision and direction of the rules to encourage good quality soccer between between evenly matched teams.

1.2 Core Vision and Requirements for Legal Standard Robot Platforms

The HSL encourages and welcomes the use of standard humanoid robot platforms available within the market to advance the state of humanoid robot soccer and the vision of the HSL.

With the HSL vision in mind, the core requirements for standard humanoid platforms used within the HSL are platforms:

1. Capable of dynamic motions such as fast walking, kicking a ball off the ground, and getting up from the ground;
2. Capable of running state-of-the-art AI neural network models for perception, decision-making, and control;
3. Sufficiently small and affordable that teams can fund multiple robots and travel with them to competitions;
4. Able to be programmed at a low-level of control;
5. Well-Documented.

Community comment, to be discussed: They make sense as ideals, but not as strict requirements. Reword §1.2 so that it is clear that these are ideals and not to be confused for the technical requirements of platforms as in §4. Maybe replace mentions to "requirements"

3 and 5 are also hard to enforce, since terms like "sufficiently small and affordable" and "well-documented" are too subjective.

1.3 Core Vision and Requirements for Constructed Robots

The HSL equally encourages and welcomes the use of fully custom built or modified humanoid robot platforms. To create a welcoming and fair environment for all robot platforms, the HSL ensures the following:

1. Both store-bought and custom-built robots can participate in a fair competition without risking damage to their robots.
2. The tournament is designed such that games are interesting for all participating teams and match-ups are fair.
3. The tournament is designed such that all currently existing teams are able to participate.
4. Details about hardware and software of the robots is made available to teams and organizers to ensure a fair competition and encourage scientific exchange.
5. League resources are distributed such that both store-bought and custom-built robots equally benefit from them.
6. Robots are designed with the goal of RoboCup in mind, thus restricting the allowed sensors where possible to humanoid sensors. Exceptions to this rule can be made if it benefits scientific progress.

2 Environment

2.1 The Field of Play

The competition consists of three distinct divisions, categorized specifically by the maximum physical height and weight of the robotic platforms.

To ensure a fair and proportionate competitive environment, the field dimensions, number of players and ball size will vary to align with the specific requirements of each division[1].

2.1.1 Field surface

The matches are played on artificial turf with a height between approximately 20 to 30 mm, except for the small division, which has a height between 8 and 12 mm, also approximately.

The color of artificial turf must be green. No particular shade is required, but the green must contrast well with the field markings and the ball and should not be very dark.

2.1.2 Field markings

The field of play must be rectangular and marked with continuous lines. These lines belong to the areas of which they are boundaries.

The color of field markings should be white, whether applied by tape, paint, or made from white turf.

The two longer boundary lines are called touchlines (or sidelines). The two shorter lines are called goalines.

The field of play is divided into two halves by a halfway line, which joins the midpoints of the two touchlines. A Team's Half is the half of the field that is closest to a team's own goal, where they primarily defend against opposing attacks.

The center mark is at the midpoint of the halfway line. A circle is marked around it, defining the center circle, a region used for kick-offs and other specific plays.

The goal area and penalty area are defined using lines drawn at right angles to the goal line.

All lines must be of the same width, which must be between 5 and 12 centimeters.

Figure 1 shows the field markings with the lines and its names.

[1]Games are played in three divisions depending on weight and height of the robots. See 3.1 for more details.

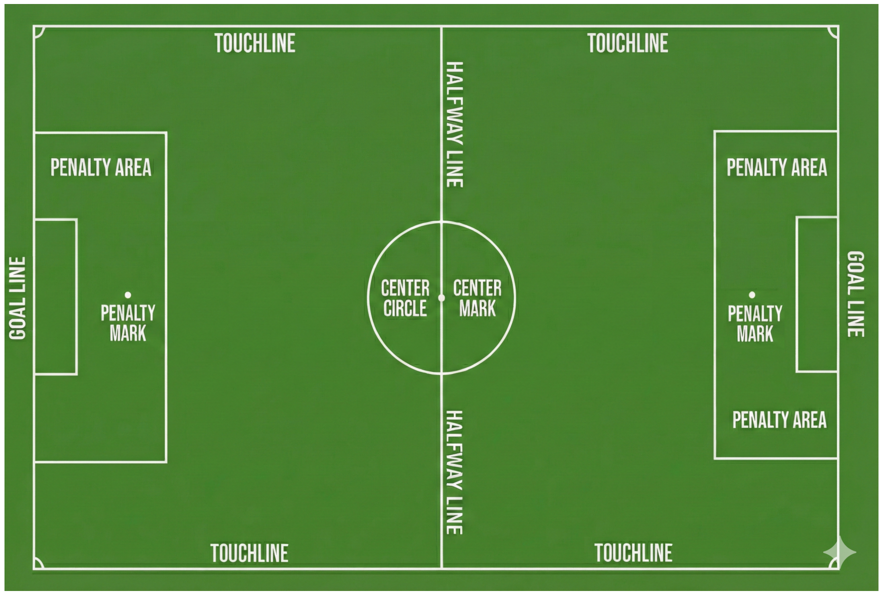

Figure 1: A labeled diagram illustrating the standard markings and key areas of a soccer field.

2.1.3 The goal area

Two lines are drawn at right angles to the goal line; the length and distance from the midpoint of the goal line depend on the division.

These lines extend into the field of play and are joined by a line drawn parallel with the goal line. The area bounded by these lines and the goal line is the goal area.

2.1.4 The penalty area

Two lines are drawn at right angles to the goal line; the length and distance from the midpoint of the goal line depend on the division.

These lines extend into the field of play and are joined by a line drawn parallel with the goal line. The area bounded by these lines and the goal line is the penalty area.

Within each penalty area, a penalty mark is made.

The distance from the midpoint between the goalposts depends on the division.

2.1.5 The corner arc

A quarter circle from each corner is drawn inside the field of play. The radius of the arc is 0.5m for the Medium Field and 1.0m for the Large Field. The Small field does not have corner arcs.

2.1.6 Dimensions

There are three field sizes that can be used for the proposed divisions, described in Table 1.

Field	Width (meters)	Lenght (meters)
Small Field (S-Field)	6	9
Medium Field (M-Field)	6 to 9	9 to 14
Large Field (L-Field)	9 to 14	14 to 22

Table 1: Approximate field sizes.

Table 2 shows the dimensions of the Small, Medium and Large soccer fields and Figures 2, 3 and 4 shows the diagram of the three soccer fields. Note that the Middle division can be played on the S-Field or the M-Field, while the Large division can be played on both the M-Field and the L-Field. More detailed technical drawings are provided in Section **??**.

Note that measurements are from the outside of the lines as the lines are part of the area they enclose (as it is in the FIFA Rules).

ID	Description (all values in meters)	S-Field (former KidSize)	M-Field (former AdultSize)	L-Field (similar to MSL)
A	Field length	9.0	14.0	22.0
B	Field width	6.0	9.0	14.0
C	Goal length (depth)	0.5 to 1.0	0.7 to 1.2	1.0 to 2.0
D	Goal width	1.8	2.4	3.0
E	Goal Area length	1.0	1.0	1.0
F	Goal Area width	3.0	4.0	5.0
G	Penalty Area length	2.0	3.0	3.5
H	Penalty Area width	4.0	6.0	7.0
I	Penalty Mark distance	1.5	2.0	2.5
J	Center Circle diameter	1.5	3.0	4.0
K	Border strip width (min.)	1.0		
L	Corner Arc radius	none	0.5 m	1.0
	Line width	0.05		0.12
	Penalty and center mark size	0.10		0.15

Table 2: Field dimensions by field type, in meters.

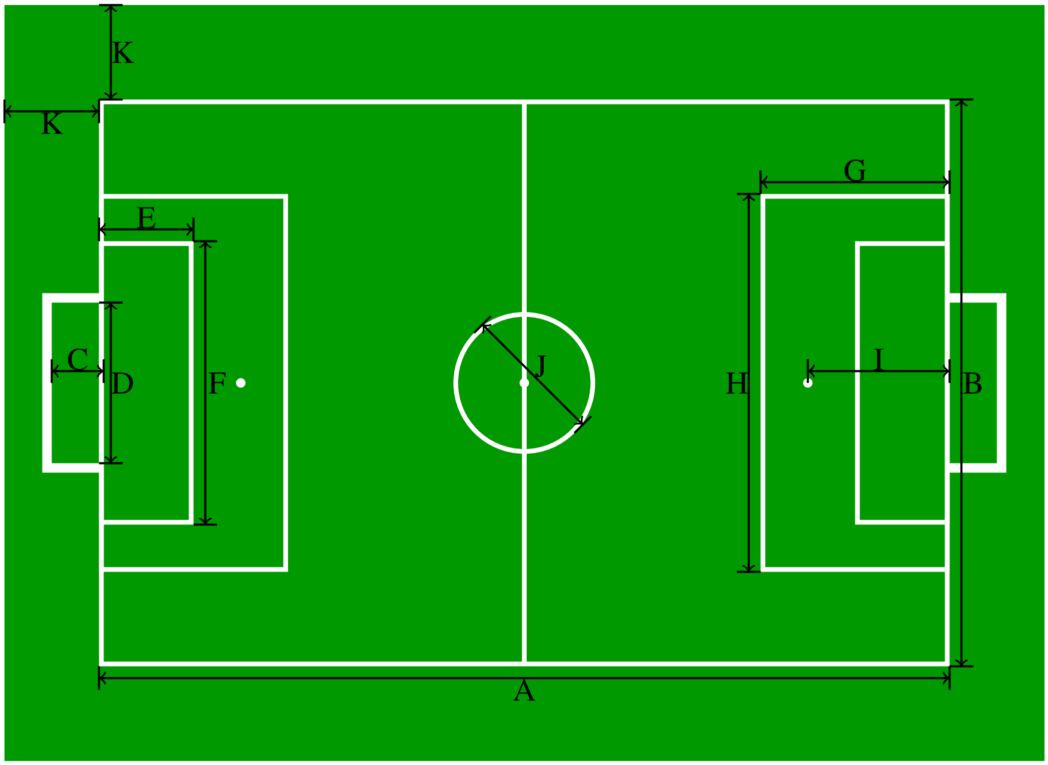

Figure 2: Schematic diagram of the small soccer field – S-Field (scale: 1/80)

2.1.7 Goals

A goal must be placed on the center of each goal line. A goal consists of two vertical posts equidistant from the corner flagposts and joined at the top by a horizontal crossbar.

The goalposts and crossbar must be made of wood, metal, or other approved material and must not be dangerous to the players.

The goalposts and crossbar of both goals must be of the same shape, which must be square, rectangular, round, elliptical, or a hybrid of these options.

need a figure how goalposts must be placed on the goal line

The goalposts and crossbar must have the same width and depth, which is not less than 8 cm and does not exceed 12 cm. The goalposts and the crossbar must be white. Goal nets and any net supports may be white, gray, or black.

Goals must be anchored securely to the ground.

Off-the-shelf goals may be used if they satisfy these requirements.

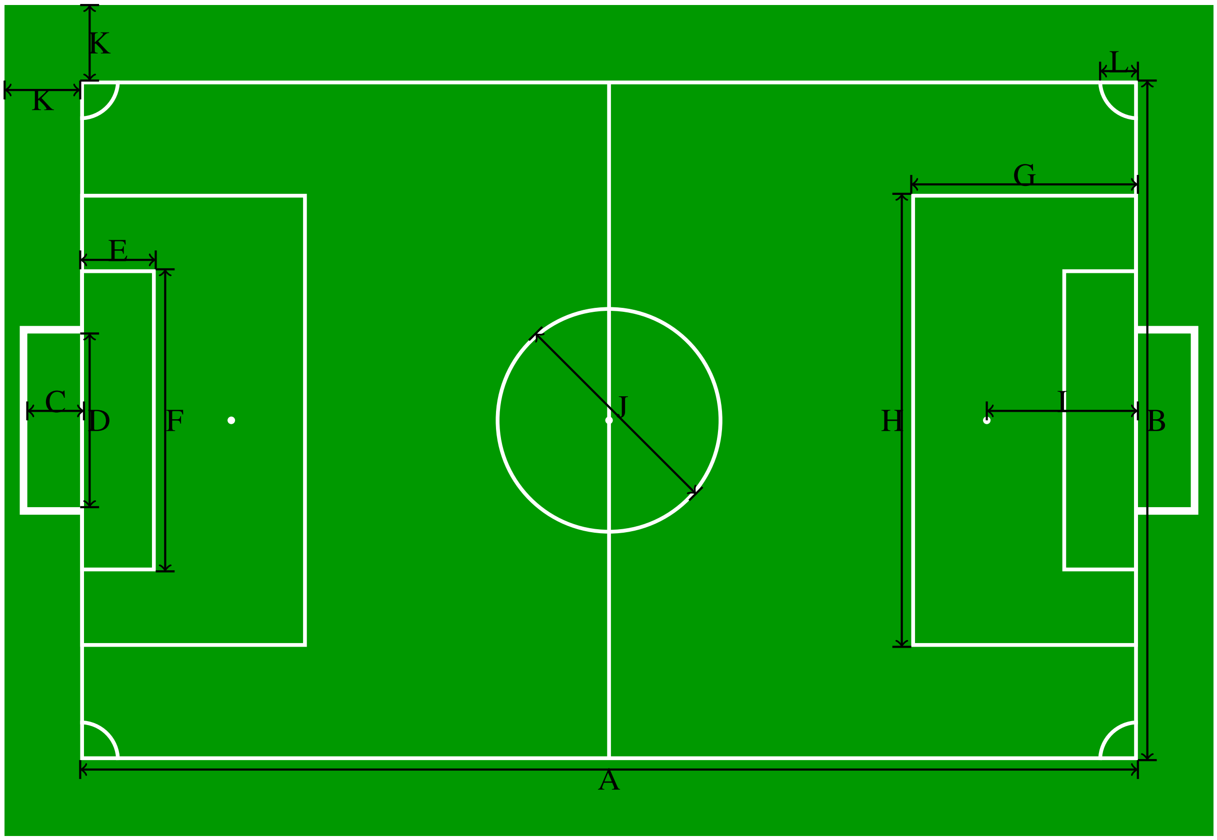

Figure 3: Schematic diagram of the medium soccer field – M-Field (scale: 1/100)

Division	Width (meters)	Height (meters)	Depth (meters)
Small	1.8	1.2	0.5 to 1.0
Middle	2.4	1.6	0.7 to 1.2
Large	3.0	2.0	1.0 to 2.0

Table 3: Approximate dimensions of the goals.

2.1.8 Definition of Inside and Outside

An object (such as a robot or the ball) is considered *inside* a region of the field if any part of it overlaps or touches the boundary lines that define that region, or if it is fully contained within the region, in the air or on the ground. It is considered *outside* the region only when no part of it, or its downwards projection, remains within or on the boundary lines of that region. This definition applies to any designated area of the field, except the goal area (See Figure **??**).

Exception for the Goal: The ball is only considered *inside* the goal (and thus a goal is scored) when the entire ball has wholly crossed the goal-side edge of the goal line, i. e., when no part of the ball remains on or above the goal line and the ball is outside the field of play.

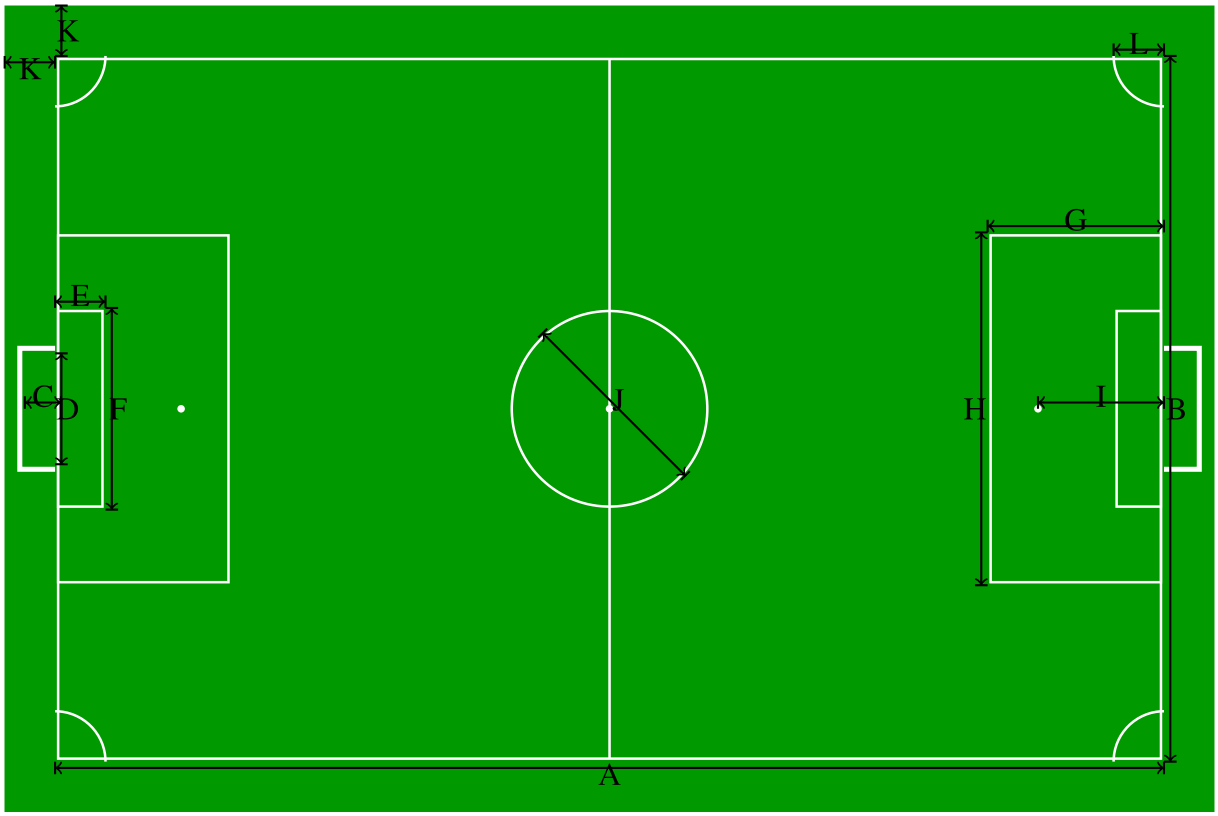

Figure 4: Schematic diagram of the large soccer field – L-Field (scale: 1/150)

2.1.9 Venue Setup

Fields may be located close to one another. Barriers will not necessarily be constructed between adjacent fields to block the robots from seeing other fields, goals, or balls. However, barriers will be constructed to block sight between any fields that are not located at least three meters apart. Hence, for each side of a field that is adjacent to another field, either barriers will separate the fields or at least $3\,\mathrm{m}$ will be between the carpet of adjacent fields.

To be decided: move venue setup and equipment requirement specific topics to an annex for local organizers?

finding from RoHOW: Barriers and Nets should be required around every field

2.1.10 Lighting Conditions

The HSL does not mandate specific or controlled lighting conditions for a match venue. It is expected that the venue provides reasonable lighting suitable for general visibility (e. g. indoor with artificial lighting, outdoor with natural lighting, or a combination of both).

The lighting conditions depend on the actual venue. Fields should be placed near or under windows where possible.

Whether window lighting is used or not, ceiling lights should be provided as necessary so that most of the field is at least $300\,\mathrm{lx}$ (preferably $400\,\mathrm{lx}$). This lighting may include variations such as glare, brightness, shadows, or mixed lighting conditions that can change throughout the match.

However, the lighting must be predominantly white, and colored lighting that significantly changes the perceived color of the field or ball is not allowed.

Teams participating in the HSL are encouraged to design their robots to handle a variety of typical lighting environments that may be encountered during a match. Natural and non-natural light must be free to reach the field. The technical committee can delimit a zone near the field where humans must not stand and where any items blocking the light sources are forbidden.

2.2 The Ball

2.2.1 Qualities and measurements

All balls must be spherical, made of or resembles the weight, form, movement characteristics and appearance of leather or other suitable material.

The balls for each division are defined in table 4.

Division	Ball Type
Small	FIFA Mini Ball [2]
Middle	FIFA size 3 or 4
Large	FIFA size 5

Table 4: Balls used in each division

2.2.2 Replacement of a defective ball

If the ball bursts or becomes defective during the course of a match: the match is stopped. It is is restarted by dropping the replacement ball at the place where the original ball became defective.

If the ball bursts or becomes defective whilst not in play at a kick-off, goal kick, corner kick, free kick, penalty kick or throw-in: the match is restarted accordingly.

If the ball becomes defective during a penalty kick or penalties (penalty shoot-out) as it moves forward and before it touches a player, crossbar or goalposts, the penalty kick is retaken.

The ball may not be changed during the match without the referee's permission.

2.2.3 Additional balls

Additional balls which meet the requirements of 2.2 may be placed around the field of play and their use is under the referee's control.

3 Teams and Players

3.1 Divisions and Configurations

A team belongs to one division, which is one of Small, Middle and Large. Main competition matches only take place between members in the same division.

Teams in each division can choose to play in two team configurations: Foundation - with a smaller number of players; and Advanced - with a larger number of players. If in a game one of the teams chooses to play in a Foundation configuration, then both teams must play in this configuration to ensure fairness. The maximum number of players in a game for each configuration is shown in Table 3.2.

In the Middle division, starting from the quarter-finals onwards, all matches will be played in the Advanced configuration.

Both the division and the configuration of a team are defined before the competition (precise modality TBD) and cannot change during the competition itself.

Maybe move some definitions from the beginning of Comp. Structure over here?

3.2 Number of Players

A match is played by two teams, each with a **maximum** number of players determined by the division the game takes place in and the configuration level of the playing teams, as illustrated by the following table:

Division	Foundation team	Advanced teams
Small	4	7
Mid	3	5
Large	3	5

At most one player per team on the field may be designated as *goalkeeper*, the others are all *field players*.

When playing at full strength, a team must have a *goalkeeper* on the field.

Goalie distinguishing feature TBD: Number? Color?

Each of the players has a unique jersey number from the set $\{1, 2, 3, \ldots, 20\}$.

Will we have flexible numbers on the game controller? Otherwise, the numbers should be limited to the game controller numbers.

3.3 Number of Substitutes

In addition, each team may prepare *substitute players* outside the field. A *substitute player* may be substituted in to become a *field player* or *goalkeeper*.

Number of allowed substitutes, needs clarity - limited, un-

3.4 Substitution procedure

3.4.1 Changing the goalkeeper

Any of the other players may change places with the goalkeeper, provided that the referee is informed before the change is made. The change is requested during a stoppage in the match.

All substitutes are subject to the authority and jurisdiction of the referee, whether called upon to play or not.

Some more sections of the Humanoid rule book, like substitution procedures for players and goalkeeper, sanctions, etc. should be moved to the chapter Game Process

4 Robot Players

4.1 The Design of the Robots

Robots participating in the HSL must have a human-like body shape with a torso, head, two arms, and two legs, as well as human-like symmetry and proportions regarding sizes of the body parts and weight distribution.

Definition of humanoid robot from Kajima et al, 2005

The robots must be able to stand upright on their feet, to walk on their legs and to be able to recover from a fall (get back to a standing position).

The only allowed modes of locomotion are bipedal walking, running, and jumping. Soccer related movements such as dribbling, kicking, or other forms of ball handling are also allowed.

The design of the robot's arms, including their length and placement, shall permit arm use and behaviors that are reasonably comparable to those of humans. Examples of permitted uses include assisting in getting up after a fall or picking up and throwing the ball (where otherwise allowed by the rules).

Arm configurations that enable behaviors significantly different from those of humans are not permitted. In particular, robots must not use their arms to provide continuous support for locomotion, such as walking on arms or using arms as additional legs.

4.1.1 Size Restrictions

All robots participating in the HSL must comply with the following restrictions:

to be discussed - including height restrictions

The length of the legs H_{leg}, including the feet, satisfies $0.35 \cdot H_{top} \leq H_{leg} \leq 0.7 \cdot H_{top}$, where H_{top} is the height of the top of the robot. The length of the leg is measured from the first rotating joint where its axis lies in the plane parallel to the standing ground to the tip of the foot.

A classic piece of human anatomy and art history, Leonardo da Vinci's "Vitruvian Man" famously depicts a man whose arm span is equal to his height, creating a 1:1 ratio. Therefore, the arm span, A_{span}, including the hands, should satisfy $0.8{\cdot}H_{top} \leq A_{span} \leq 1.2{\cdot}H_{top}$.

Based on H_{top}, the following size restrictions apply for each division:

- Small: $H_{top} \leq 1.1$ meters;
- Middle: $H_{top} \leq 1.25$ meters;
- Large: $H_{top} \leq 1.9$ meters.

H_{top} is defined as the height of the robot when standing upright (with fully extended knees). H_{top} is measured with the head of the robot oriented in such a way that it is tilted to either its maximum upwards tilt angle or the horizon line, whichever is lower.

The height of the head H_{head}, including the neck, satisfies $0.1 \cdot H_{top} \leq H_{head} \leq 0.3 \cdot H_{top}$. H_{head} is defined as the vertical distance from the axis of the first arm joint at the shoulder to the top of the head.

4.1.2 Weight Restrictions

The robot's Body-Mass Index (BMI) is defined as follows: $\mathrm{BMI} = \frac{M}{{H_{top}}^2}$, where M is the mass of the robot in kg and H_{top} its height in meters.

The Body Mass Index (BMI) of the robot should be: $5 \leq \mathrm{BMI} \leq 30$.

The following weight restrictions apply for each division:

- Small: Weight$\leq$ 15 kilograms;
- Middle: Weight $\leq$ 25 kilograms;
- Large: Weight $\leq$ 80 kilograms.

4.1.3 Safety

A player must not use equipment or wear anything that is dangerous to himself or another player.

From the humanoid league rules, to be discussed

Robots competing in the physical competition must be equipped with an emergency stop button that makes the robot immediately desist with all motions, or ideally go limp and/or cut power to the actuators. In addition to the emergency stop button, robots may only have up to two additional physical or virtual buttons: One to start the robot behaviour and one to stop the behaviour. The buttons must be clearly labeled. If the robot has more buttons that cannot be detached, they must be visibly masked during the games.

to be discussed: 4.7.1 allows buttons, community also re- ... the game can be restricted

In ...tbd... size, robot handlers are allowed to carry an additional remote emergency stop button. This button must be worn either around the neck or on the belt of the robot handler and must be clearly marked. Each emergency stop button can only be connected to the robot of the robot handler that holds the button. The remote emergency button cannot perform any additional functions and does not replace the regular emergency button. Robot handlers must keep their hands clearly away from the button unless the button is being pressed. Robot handlers must not use the remote emergency button to intentionally incapacitate their robots.

to be discussed, will there still be robot handlers?

to be discussed, add other safety measures from our discussion?

4.2 Approved Standard Platforms

The following commercially available robotic platforms are approved for participation in the HSL.

The following are the list of pre-approved standard platforms that require no modifications:

Manufacturer	Model	Restrictions
Aldebaran	Nao V5, and Nao V6	
Robotis	DARwIn-OP	
Booster	T1, and K1	
Fourier	GR1, GR1-Pro, and GR2	
Unitiree	G1++, H1++	

Additional humanoid robot platforms can be approved upon request. Please send your request for approval to `rc-spl-tc@lists.robocup.org`.

Change to new mailing list when ready

4.3 Hardware

to be discussed, divisions, list of permitted robots, self-built robots, etc.

Modifications or additions to the robot hardware are allowed.

to be discussed

No additional hardware is permitted, including off-board sensing or processing systems. Additional sensors besides those originally installed on the robots are likewise not allowed.

4.4 Sensors

Teams participating in the HSL competitions are encouraged to equip their robots with sensors that have an equivalent in human senses. These sensors must be placed at a position roughly equivalent to the location of the human's biological sensors. In particular, ... to be discussed

The sensors and their placement shall be chosen such that they allow the robot a spatial perception similar to humans. The sensors are evaluated along the following two general guidelines:

1. *Foster and encourage research and development towards human-like perception capabilities.* Sensors aiming to directly emulate human senses, like a camera or a microphone. Use of such sensors is explicitly *encouraged*.

2. *Enable research and development under the constraints of the current state of the art in technology and research, as long as this does not undermine current research efforts as declared in point 1.*
   Sensors that compensate for current shortcomings in technology and state of research, like one-dimensional distance sensors with limited range, or two vertically arranged cameras in the robot NAO.

Generally, *passive sensors* are preferred to *active* sensors that actively emit signals.

4.4.1 Intrinsic Perception and Proprioception

Any sensing capability aiming at measuring the internal state of the robot is permitted. This includes temperature, current consumed by the motors, joint positions, etc.

4.4.2 Visual Perception

The visual sensors, e.g., cameras of the robot shall be arranged such that the combined visual field is *contiguous* and limited to dimensions similar to a human, which corresponds to limitations in the opening angle: horizontal $\leq 180°$ and vertical $\leq 140°$.

The visual sensors shall be located in the head of the robot.

The combined dynamic visual field that can be observed by the robot solely by moving its cameras (similar to human eye movements) and the head is limited to horizontal $\leq 340°$ and vertical $\leq 220°$.

The cameras are restricted to visual information in the range of the light visible to humans.

The cameras can provide dense visual information, such as a rasterized image, or sparse information, such as visual events (event cameras).

4.4.3 Visual (dense) Depth Perception

Passive integrated devices that provide dense depth information, such as stereo cameras, are permitted and encouraged.

Active integrated devices that provide depth information, such as cameras with an active infrared projector or time-of-flight cameras, are permitted but discouraged. Their use might be prohibited in the future.

4.4.4 Orientation

Sensors providing information regarding the robot's orientation in space with relation to the ground are permitted. This includes sensors such as gyrometer, accelerometer, as well as integrated inertial measurement units (IMU), as long as they provide only relative measurements and no measurements of the absolute direction, such as a compass.

An IMU with an integrated compass can be used as long as the compass is disabled.

The sensors can be placed in the head and/or in the torso of the robot.

4.4.5 Sound Perception

The sound sensors, e.g., microphones, shall be placed in the head of the robot. The number of microphones is limited to ??

4.4.6 Haptic Sensing

Any passive sensor allowing haptic measurement is permitted, such as force sensors, touch sensors, buttons/bumpers, capacitative touch sensors.

Haptic sensors can be placed at any location of the robot's body and are not limited in number.

4.4.7 Distance Sensing

Active and passive sensors for one-dimensional distance sensing with a limited range are permitted if their use is limited to compensate for shortcomings in the spatial awareness of the robot in the close proximity.

The number of such distance sensors is limited to 4.

The use of distance sensors is discouraged and might be prohibited in the future.

4.4.8 Summary

The following table briefly summarizes pre-approved sensors. The use of listed sensors that are considered human-like is encouraged. The use of listed sensors that are not considered human-like is accepted, but discouraged and might be prohibited or limited in the future.

Sensor Type	Human-Like	Comments
RGB camera	yes	opening angle limit: horizontal $\leq 180°$, vertical $\leq 140°$
Stereo camera	yes	
Event camera	yes	
Active RGB-D based on infrared projection	no	discouraged
Active RGB-D based on time of flight (TOF)	no	discouraged
Microphone array	yes	
Gyro	yes	
Accelerometer	yes	
Compass	no	magnetic measurement sensors are not allowed
Integrated inertial measurement units (IMU)	yes	no compass or must be disabled and not used
Force sensors	yes	
Touch sensors	yes	
Buttons / Bumpers	yes	
Capacitive touch sensors	yes	
Near range infrared sensor	no	
Sonar / ultrasound sensors	no	
1D LIDAR and laser range sensor	no	with limited range
Intrinsic sensors	yes	temperature, current, joint positions, etc.

Reinaldo: I think we forgot about compass, so I added to the table

Clarity needed for manufacturers that do not provide IMU details, just access to data

4.5 Team Markers

Robot bodies must be colored mostly with a neutral color, such as black, gray, white, or silver, and be non-reflective.

Robots must use colored jersey shirts as team markers. A jersey is a tank top-style shirt that may start at the neck and go down to the waistline, and does not cover the arms of the robots. It must cover at least 50% of the upper body of the robot.

Jerseys must have a primary color that comprises more than half of the jersey. The primary color must be recognizable from the front and back.

Jerseys should not contain distracting patterns that could be confused with other elements of the soccer field, such as field lines or the ball.

Jersey material must be non-reflecting, non-shiny, and non-glittery. Jerseys may be manufactured from fabric and fine mesh.

The two teams must wear colors that distinguish them from each other and also the referee and the assistant referees.

Each team must provide two jersey designs in distinctly different solid colors. The jersey color must clearly contrast with any uncovered parts of the robot to ensure reliable identification.

Robots must display clear identification numbers (player number): a large number on the back and a smaller number on the front.

The goal keeper robot must be marked uniquely so that it can be easily distinguished from the other robots of a team by the referees.

Sponsorship logos will be allowed on jerseys, though detailed guidelines are still being finalized.

Jersey designs must be submitted for approval prior to the competition. to be discussed

4.6 Goalkeeper

The *goalkeeper* may use any of the allowed jersey numbers. The *goalkeeper* must wear a jersey with a primary color different from the primary colors used by the *field players* of both teams.

4.7 Communication

Robots participating in the HSL competitions must act autonomously during a game.

Any use external power supply, teleoperation, remote control, remote processing, remote sensing, or remote brain of any kind is prohibited.

Communication is only allowed among robots on the field, and between the robots and the Game-Controller. between the robots and the referees,

4.7.1 Acoustic and Visual Communication

The robots are permitted to communicate through visual signaling (e. g., gestures) and / or acoustic signaling using the robot's built-in microphones and speakers.

Communication that causes excessive discomfort to an audience, affects the safety of an audience, or violates normal playing rules is not permitted.

4.7.2 Wireless Communications

The robots are permitted to communicate through WiFi using builtin wireless adapters.

A field is equipped with a wireless access point that must be used for any wireless communication during the game. Only the teams involved in the game are permitted to use the access point during a match.

Each team will be assigned a range of IP addresses that can be used both for their robots and their computers. The network configuration (e. g. IP addresses, channels, SSIDs, and required encryption) of the fields will be announced at the competition site.

4.7.2.1 Robot-To-Robot Communication: Wireless robot-to-robot communication between the robot players is allowed and must the access points provided by the event organizers. Direct wireless communication between the robots (using the so-called ad-hoc mode) is prohibited.

Messages must be sent via UDP broadcast. Each message's payload size must not exceed 512 Bytes.

Each team will be allocated a single UDP port on which to send/receive their team messages. Specifically, a team's port will be 10000 plus that team's GameController number.

Unicast communication between robots is prohibited.

Each team is allowed to send in total a maximum of 12000 messages during a complete game.

For each minute of extra game time, the limit is increased by 600 packets.

Only messages sent by the robot players in the following game states count towards the communication limit: ready, set and playing. Any messages sent during other game states do not count towards the teams message budget. Limits do not apply during penalty kick shoot-out.

The number and the size of the messages sent by each team is monitored by the GameController.

A violation of the permitted maximal size of a message or the maximal number of messages is subject to penalty.

What is the penalty?

The limit of allowed packets will be lowered in future competitions to encourage a progress towards human-like communication and coordination.

4.7.2.2 GameController Communication: Robots are permitted to send status update packets to the GameController.

describe GC communication

4.7.2.3 Debug Communication: Team specific debug information may be sent to an external computer owned by the team. A robot may send debug information at most once every ??? in a single UDP packet.

5 Law 5 – The Human Officials

Each match is run by a number of human officials. They are distinguished in two categories: referees, neutral persons who have the the duty to enforce the rules of play and ensure the game proceeds smoothly (including the GameController operator); and team officials, who belong to either of the teams and perform specific tasks for that team.

The referees are assigned before the game: example modalities are by scheduling in official games or by common agreement in friendly games.

As for team officials, each team selects theirs officials before the start of the game and communicates their selection to the head referee during the pre-game meeting. One person may cover multiple roles, unless otherwise specified.

5.1 Law 5.1 - The head referee

The head referee is the principal authority of the game and takes all decisions regarding the flow of the match and the enforcement of the rules.

5.1.1 Decisions of the head referee

The head referee takes decisions to the best of their ability according to the rules and the spirit of the game. Any decision of the head referee is valid.

The head referee's decision is final and can not be changed afterwards, even by video proof. There is no discussion about decisions during the game, neither between the assistant referees and the head referee, nor between the audience or the teams and the head referee.

The head referee cmmunicates their decision via verbal calls, occasionally complemented by a whistle sound.

Calls. The head referee announces decisions by a clear loud call, and (as required) whistle sound. The whistle, or where there is no whistle the first verbal word of the referees calls, defines the point in time at which the decision is made. The referees should make efforts to use consistent and clear calls, and it is preferable for referees to use the calls as specified in these rules.[3] The intention of specifying the referee calls is for clarity and consistency across games.

Whistle. Where a whistle is required, the head referee first whistles and then announces the reason for the whistle. The head referee may choose to use any normal sports whistle. Each whistle sound should be short and not too loud as to interfere with other fields and simultaneous games. The head referee must only sound the whistle in circumstances described in these rules.

[3]The calls specified in these rules are detailed in English. With the agreement of the teams, the referees may use suitable calls in any language. The exception to this are technical challenges that depend on the calls as specified.

5.1.2 Powers and duties

The head referee enforces the rules of the game and controls the match in cooperation with the assistant referees and the GameController operator.

The head referee must ensure that the equipment worn by all robots and humans on the field meets the relevant requirements (Laws ???).

The head referee adjudicates all infringements of the rules, and stops, suspends, or abandons the match accordingly. They indicate the restart of play after it has been stopped.

This is the "advantage rule", decide if we want it or not

The head referee may allow play to continue when the team against which an offence has been committed will benefit from such an advantage, and penalises the original offence if the anticipated advantage does not actually ensue.

When a robot commits multiple offenses at the same time, only the most serious one is called.

The head referee should act on the advice of the assistant referee regarding incidents that they has not seen.

The head referee has the right to take action against team officials who fail to conduct themselves in a responsible manner and may, at their discretion, expel them from the field of play and its immediate surrounds.

The head referee must stop the game if a human is seriously injured during play until that person has left the field.

The head referee can stop, suspend, or abandon the match because of outside interference of any kind, such as but not limited to: poor lighting or network conditions, or extraneous objects, animals or people entering the field.

The head referee can request any robot be stopped if it becomes a threat to the safety of objects, participants or spectators.

The head referee ensures that no unauthorised persons enter the field of play.

The head referee should avoid handling the ball (except for placing a ball for kick-off), and avoid handling the robots. Their duty is to monitor and adjudicate the game. The head referee should only handle robots and the ball if absolutely necessary to expedite gameplay or removal of penalized robots, where the assistant referees are otherwise occupied or too far away, and only if it doesn't compromise their own safety.

The head referee may decide at any point before or during a game to relocate any objects around the field, or direct persons to another position around the field.

5.2 Law 5.2 - Assistant Referees

The assistant referees assist the head referee with offenses and other events when they have a clearer view. For example, they indicate when the ball has completely crossed over the line of the field, and which team is awarded the resulting goal or free kick. They can also point out rule violations that occurred outside of the head referee's view.

The assistant referees are also tasked with placing the ball for a free kick after it goes out of the field.

Note that only calls made by the head referee are valid. Still, they should take the assistant referees' input into account, especially when the assistant was in a better position to observe the event.

A game normally has one or two assistant referees. The exact number is decided by the organizers of the competition.

5.3 Law 5.3 - GameController Operator

The operator of the GameController is a referee who sits at a PC in the technical area. As with the head referee, the operator should make efforts to use consistent and clear calls.

They will signal any change in the game state or penalties to the robots via the wireless as they are announced by the head referee. Note that for both kick-offs and goals, the moment of whistling is determining, not the verbal announcement of the head referee. They should repeat the call of the head referee as they do so, to confirm it was heard correctly.

The operator will also inform the robot handlers when a timed penalty is over and a robot has to be placed back on the field. They should announce events that occur automatically in the GameController due to elapsed time, such as the ball coming into play after a kick-off, penalty kick or free kick, or the state changing from `ready` to `set`.

If the penalty procedure would benefit from an advance warning about 10 seconds before the end of the penalty, it can be mentioned here.

They are also responsible for keeping the time of each half. They should count aloud the remaining seconds in a half once the time remaining is 5 s or less.

5.4 Law 5.4 - Equipment of the referees

All referees should wear a suitable referee jersey, and black or dark-blue socks, and they must avoid reserved colors (white and green) in their clothing.

The head referee must also be equipped with a whistle, a coin, and yellow and red cards.

The referees may also have equipment to communicate with each other, such as a headset or flags, if available.

5.5 Law 5.5 - Referee List for Friendly Games

During a competition, especially (but not only) during the setup days, several teams may want to participate in friendly games with each other if a field is available to play in. People willing to volunteer for judging these games as head referee, assistant referee or GameController operator may submit their name to a list managed by the Organizing Committee, so that the teams organizing the friendly game are aware of their availability. This is especially recommended for those who wish to gain referee experience.

Ultimately, the teams organizing the friendly game are still free to decide whether to call volunteers from the list or otherwise choose their referees.

The Organizing Committee is in charge of maintaining the referee list and should be approached at the competition site if one should want to volunteer.

5.6 Law 5.6 - Team Captain

The captain is a team official. They are the representative of the team for the game. They do not have to be the team leader.

The team captain represents the team during the pre-game meeting.

The team captain is the only person allowed to communicate with the referees. This includes making requests for pickup (Section ???).

5.7 Law 5.7 - Robot Handlers

The robot handlers are team officials. They are the only people allowed to touch robots of their own teams (except for those that are picked up). Each team must provide one or two handlers.

During the game, the robot handlers stay in a designated area and must receive permission from the head referee before entering the field.

The robot handlers are tasked with removing their robots from the field when they are penalized, picked up, or otherwise removed from play. They must do so as swiftly as possible when instructed by the head referee, and with minimum interference with the game.

Put either here or in forbidden actions: refusing to follow head ref's instructions is considered unsporting conduct

When play is stopped, the robot handlers inform the head referee when their robots are ready to resume.

Robot handlers should only enter the field to execute a decision made by the head referee. They should not prevent robots from falling during an official game.

A robot handler may not touch a robot from a different team.

5.8 Law 5.8 - Humans allowed on the field

During the game, starting at the Ready state, only the following people are allowed on the carpeted area (i.e. on the field and the surrounding border area):

- The referees;
- The robot handlers;
- And no one else.

In addition, only the team captains and at most two more people per team are allowed next to the GameController tables. The remaining humans locate themselves around the other sides of the field if they want to watch the match. This allows the referees easier communication with the team and the GameController operator gets less disturbed.

The exact name of this area depends on how the tables are laid out

6 Law 6 – Judgement

6.1 Law 6.1 - Pre-game Team Meeting

Law 7.x

Both teams send a representative called team captain to the field 10 min before their match starts. This time should be used to welcome each other, assign team colors (see Section ???), choose side and kick-off (see Section ???), and discuss any other topics related to the match.

6.2 Law 6.2 - Referee–Team Communication

During the match, only the team captains are allowed to communicate with the head referee.

After the match the teams thank the referees for their duty.

During all phases of the match teams and referees are communicating with respect to each other.

6.3 Law 6.3 - Referees during the Match

The head referee and the assistant referees usually oversee the game from outside the field. They may enter the field in particular situations, e. g., to remove a robot when applying a penalty. They should avoid interfering with the robots as much as possible.

Pending safety for bigger divisions

6.4 Law 6.4 - Referees after the match

At the end of the game, the head referee must provide the Technical and Organizing Committees (or other relevant authorities) with a match report, which includes the final score and any relevant information on incidents that occurred before, during or after the match.

7 Game Process

Law 7 and the following ones.

8 Law 7 – The Duration of the Match

The match consists of three parts: the first play period (first half), a half-time break, and a second play period (second half).

8.0.1 Periods of play

Each game period has a duration of 10 minutes.

8.0.2 Half-Time Break

The half-time break must not exceed 10 minutes.

During the half-time break both teams may perform any modifications on the hardware or software of the robots, change robots, or do anything else that can be done within the time allotted.

8.0.3 Signals

The head referee signals the beginning of each half with a single whistle blow. The head referee signals the end of the first half with two short whistle blows, and the end of the second half with two short plus one long whistle blow. The head referee should make all whistle sounds from the T-junction of the halfway line.

8.0.4 Interruptions

round robin - no interruption of the time + game can end in a draw

elimination - interruption of the time on ready, set, ... + game time prolongs for the duration of potential penalty - or is it a separate game phase?

8.0.5 Penalty Kick

If a penalty kick has to be taken or retaken, the duration of either half is extended until the penalty kick is completed.

8.0.6 Set and Ready states

During the Set and Ready states, the game clock should not be stopped in both knock-out and round-robin games.

8.0.7 Allowance for time lost

Allowance is made in either period for all time lost through:

- substitutions
- assessment of injury to players
- removal of injured players from the field of play for treatment
- wasting time
- any other cause

The allowance for time lost is at the discretion of the referee.

9 Law 8 – The Start and Restart of Play

A kick-off starts both halves of a match, both halves of extra time and restarts play after a goal has been scored. Free kicks (direct or indirect), penalty kicks, throw-ins, a goal kicks and corner kicks are other restarts (see Laws 13–17). A dropped ball is the restart when the referee stops play and the Law does not require one of the above restarts.

If an offence occurs when the ball is not in play, this does not change how play is restarted.

9.0.1 Kick-off

Procedure

- The referee tosses a coin and the team that wins the toss decides which goal to attack in the first half or to take the kick-off.
- Depending on the above, their opponents take the kick-off or decide which goal to attack in the first half.
- The team that decided which goal to attack in the first half takes the kick-off to start the second half.
- For the second half, the teams change ends and attack the opposite goals. The kick-off at the beginning of the second half has to be taken by the team which did not take the kick-off in the beginning of the first half.
- After a team scores a goal, a kick-off has to be taken by their opponents.

For every kick-off:

- all players must be in their own half of the field of play,
- the opponents of the team taking the kick-off must be outside from the center circle until ball is in play,
- the ball must be stationary on the center mark,
- the referee gives a signal,
- the ball is in play when it is kicked and clearly moves,
- if the ball does not move 10 seconds after the referee has given the signal then the ball is in play and the opponents of the team taking the kick-off may enter into the center circle,

From humanoid league, to be discussed

- a goal may not be scored directly against the opponents from the kick-off,
- if the team taking the kick-off has three or more robots on the field then two different robots need to touch the ball before scoring a goal. If a team taking the kick-off has only two or less robots on the field, the robot taking the kick-off has to touch the ball at least one time outside the center circle before scoring a goal.

This statement needs confirmation by TC

Offences and sanctions

In the event of any kick-off procedure offence, the kick-off is retaken.

9.0.2 Dropped ball

Definition of dropped ball

A dropped ball is a method of restarting play when, while the ball is still in play, the referee is required to stop play temporarily for any reason not mentioned elsewhere in the Laws of the Game. In the virtual competition, the only reason for a dropped ball to be called is that the ball has moved less than 5 centimeters in the last 2 minutes of play.

Dropped ball definition and procedure are different in FIFA rules and in RC-HL rules. RC-HL rules are used in current edition because FIFA rules of dropped ball are hardly to be implemented by robots

Procedure

The game is continued at the center mark. A goal can be scored directly from a dropped ball. The procedure for dropped ball is the same as for kick-off, except that the players of both teams must be outside the center circle. The ball is in play immediately after the referee gives the signal

Offences and sanctions

- If a player moves too close to the ball before the referee gives the signal, a kick-off is awarded to the opponent team.
- The ball is dropped again:
  - if it is touched by a player before it makes contact with the ground,
  - if the ball leaves the field of play after it makes contact with the ground, without a player touching it.

10 Law 9 – The Ball In and Out of Play

10.0.1 Ball out of play

The ball is out of play when:

- it has wholly crossed the goal line or touch line whether on the ground or in the air
- play has been stopped by the referee

10.0.2 Ball in play

The ball is in play at all other times, including when:

- it rebounds off a match official, goalpost, crossbar or corner flagpost and remains in the field of play

11 Law 10 – The Method of Scoring

11.1 Goal scored

A goal is scored when the entire ball (not only the center of the ball) passes over the goal line, between the goalposts and under the crossbar, provided that no infringement of the Laws of the Game has been committed previously by the team scoring the goal.

The goal-point is awarded to the opponent team of the goal that was scored into.

Own goals can be scored even if there was an infringement?

The restart of the play will be a goal kick for the opponents team.

11.1.1 Winning Team

The team scoring the greater number of goals during a match is the winner. If both teams score an equal number of goals, or if no goals are scored, the match is drawn.

11.1.2 Competition Rules

When competition rules require there to be a winning team after a match or home-and-away tie, the only permitted procedures for determining the winning team are those approved by the International F.A. Board, namely:

- extra time
- kicks from the penalty mark

12 Law 11 – Offsides

Offsides are not used in the RoboCup Humanoid Soccer League.

13 Law 12 – Fouls and Misconduct

13.1 Procedure

The referee has the authority to take disciplinary sanctions from the moment they enter the field of play until they leave the field of play after the final whistle (including penalty shootouts). Removal penalties, warnings, yellow and red cards are used to discipline players according to the nature of the offense committed. Warnings, yellow and red cards given to robots only accumulate for the current game and are cleared after the end of each game.

13.2 Removal Penalty

Robots committing offenses must be removed from play and must serve a time penalty of 30 seconds. A removal penalty is issued for the following offenses:

- Being incapable of play as defined below
- Being granted a pick up as defined below
- Holding the ball for more than 5 seconds as defined below
- Receiving a warning or yellow card
- Any other offense not further defined by the rules

A removal penalty is issued by the head referee announcing ”Penalty [team color] [player number]”.When a penalty is called, the designated robot handler has to remove the robot immediately and by that interacting as little as possible with the game. While the robot is removed from play, the team may perform any kind of maintenance on the robot. However, the robot may only reenter the field after maintenance is complete and will only start serving its penalty after it has been placed. When the robot is ready to serve its penalty, it is placed on the sideline of the field on the height of the penalty mark of the own team, facing the opposing sideline. After the robot handler has let go of the robot, they announce ”[team color][player number] ready”. The referee acknowledges this and the robot starts serving its time penalty.

If the robot re-enters the field before its time penalty has fully elapsed, or if it is touched by a human after it has begun serving the penalty, the robot must restart its 30 second time penalty and receives a warning.

If a removal penalty occurs within 1 m of the ball, play must be stopped. Play is resumed via a free kick.

13.3 Definition of Incapable Robots

Players not capable of play (e.g. players not walking on two legs, players not able to stand, or players with obvious malfunctions) are not permitted to participate in the game. They must be removed from the field. It is up to the referee to judge whether a player is capable of play. The referee may ask the team leader of a player suspected to be incapable of play to demonstrate playing ability at any time. A player is incapable of play if

- The player is obviously damaged
- The player fell and is not able to get back into a standing or walking position within 20 seconds
- The player is the goalkeeper, a ball is within 0.5 m of the robot within the goal area, and the robot does not show any attempt to remove the ball
- The player poses a threat to itself, other players or humans
- The player is damaging the field or the arena

13.4 Request for Pick-up

A robot handler may request to pick up a robot if and only if a robot is in a dangerous situation that is likely to lead to physical injuries. The head referee must approve this request prior to the robot handler touching the robot. If a team member touches a robot without the allowance of the referee, the respective robot and the person touching it each receive an official warning. The referee may deny a request for pick up. A pick up is not granted for reasons such as being incorrectly configured or being unable to see the ball.

13.5 Illegal positioning

A robot is deemed to be illegally positioned if, during the SET phase, it is located in the opposing half, outside the field of play, or inside the centre circle when it does not have kick-off. Robots that are illegally positioned must be removed from play and placed on the sideline as described in the removal penalty section. However, unlike a removal penalty, an illegally positioned robot does not incur a time penalty and may re-enter the field immediately once play has resumed.

Add approaching free kicks too?

13.6 Ball Holding

A player receives a removal penalty if it holds the ball for more than 5 seconds in a way that the ball cannot be removed from the player. The goal keeper may hold the ball for up to 10 seconds. To be considered removable, more than half of the ball's volume must be outside the convex hull of the player, projected to the ground. If the ball enters the convex hull repeatedly, it must be removable in between for the majority of the time. If more than one player of a team is in the vicinity of the ball, the convex hull is taken around all players of the same team who prevent removal of the ball.

13.7 Untangling

If entangled robots fail to untangle themselves, the referee might ask designated robot handlers of both teams to untangle the robots. Untangling must not make significant changes to robot positions or heading directions. Untangled robots must be laid on the ground not closer than 50cm to the ball and in a way not gaining an advantage. Being entangled does not incur a penalty.

13.8 Cautionable offenses

The yellow card is used to communicate that a player has been cautioned. A player is cautioned and shown the yellow card if they commit any of the following offenses:

- Unsporting behaviour
- Pushing as defined below
- Damaging the field of play or the arena
- Persistent infringement of the Laws of the Game
- Delaying the restart of play
- Entering the field of play without the referee's permission
- Receiving a second warning

13.9 Definition of Pushing

A robot is cautioned for forcefully pushing an opposing player in a manner that significantly destabilizes the player. Examples for forceful contacts include falling into another player or walking into another player at significant speed. A robot is also cautioned if it walks into another player for more than 5 seconds, even when the applied force is minimal.

Front to front contact between players does not lead to a caution unless one player walks with a significantly higher speed or with significantly more force. A player that is currently getting up from a fall cannot commit a pushing offense.

Should kicking a player also be punished?

13.10 Sending-off offenses

The red card is used to communicate that a player has been sent off. A player is sent off if they commit any of the following offenses:

- Serious foul play
- Endangering any human
- Receiving a second yellow card

A player who has been sent off must leave the vicinity of the field of play and the technical area for the remainder of the game.

13.11 Substituting a Cautioned Player

Teams may substitute a player who has received official warnings or cautions as usual. However, all warnings and cautions issued to the substituted player are transferred to the player entering the field. Players who have received a red card may not be substituted. As a result, the team must play the remainder of the match with one fewer player.

13.12 Penalties against Teams and Humans

The referee can issue yellow and red cards against humans or teams at their discretion. Cards can be issued for offenses such as

- Offensive, violent or abusive behaviour

- Repeatedly ignoring referee instructions
- Repeatedly entering the field without permission
- Breaking the game rules to gain an advantage

Cards that are issued against humans or teams must be reported to the TC immediately after the game. Cards against humans or teams persist between games. Receiving two yellow cards results in a red card. If a human receives a red card, they will not be allowed to enter the field or team area during games for the remainder of the competition. If a team receives a red card, they are disqualified from the competition.

14 Free Kick

A free kick is a method of restarting play after various stoppages.

Discuss: Maintain indirect + direct free kicks, AND indirect kick rule?

A free kick is initiated in the following situations:

- When the ball goes over the touchlines, termed *Kick-in* or *Throw-in* (see Section 15).
- When the ball goes over the goal line having last touched a player of the defending team, termed *Corner Kick* (see Section 17).
- When the ball goes over the goal line having last touched a player of the attacking team, termed *Goal Kick* (see Section 16).
- When a pushing foul or other infringement is awarded (see Section 13.9), termed a *Pushing Free Kick*.

The head referee will announce a free kick by calling one of:

1. "Kick-in <color>" / "Throw-in <color>" for the team awarded the kick-in.
2. "Corner Kick <color>" for the team awarded the corner kick.
3. "Goal Kick <color>" for the team awarded the goal kick.
4. "Foul <offending color><offending number>" for pushing free kicks.

The GameController will then activate the sub-state for the respective free kick. The team awarded the free kick (termed the *attacking team*) has 30 s to complete the kick. Free kicks are classified as either **direct** or **indirect** (see Sections 14.1 and 14.2).

14.1 Direct Free Kick

A direct free kick allows the attacking team to score a goal directly from the kick:

- If a direct free kick is kicked directly into the opponent's goal by the attacking team, a goal is awarded.
- If a direct free kick is kicked directly into the team's own goal, a corner kick is awarded to the opposing team.

The following set plays are direct free kicks: *Corner Kick*, *Goal Kick*, and *Pushing Free Kick*.

Note: If the attacking team fails to execute the free kick within the time limit, the defending team may also score directly (see Failed Free Kick in Section 14.3).

14.2 Indirect Free Kick

An indirect free kick requires the ball to be touched by another player before the attacking team can score a goal:

- A goal can only be scored by the attacking team if the ball has been touched by another player (of either team) after the kick, before being kicked again into the goal.
- If an indirect free kick is kicked directly into the opponent's goal by the attacking team without touching another player, a goal kick is awarded to the opposing team.
- If an indirect free kick is kicked directly into the team's own goal, a corner kick is awarded to the opposing team.

The following set plays are indirect free kicks: *Kick-in* / *Throw-in*.

Note: If the attacking team fails to execute the free kick within the time limit, the defending team may score directly without the indirect requirement (see Failed Free Kick in Section 14.3).

14.3 Execution

The referee places the ball according to the type of free kick (see Sections 15 to 17 for specific ball placement rules). For a *Pushing Free Kick*, the ball is left in place, repositioned only if required by the pushing rules (see Section 13.9).

14.3.1 Ball Position and Placement

All free kicks are taken from the place where the offence occurred, except:

- Indirect free kicks to the attacking team for an offence inside the opponents' penalty area are taken from the nearest point on the penalty area line which runs parallel to the goal line.
- Free kicks to the defending team in their goal area may be taken from anywhere in that area.

The ball must be stationary and is in play when it is kicked and clearly moves as determined by the referee, except for a free kick to the defending team in their penalty area where the ball is in play when it is kicked directly out of the penalty area. The ball is also considered in play 10 seconds after the referee gave the signal.

14.3.2 Avoidance Region and Distance Requirements

During a free kick, only the attacking team may approach within the avoidance region of the ball. For a goal kick, the avoidance region is the entire penalty area. For all other free kicks, the avoidance region is division dependent: $0.75\,\mathrm{m}$ (Small), $1.0\,\mathrm{m}$ (Medium), $1.5\,\mathrm{m}$ (Large).

Should be the whole penalty area for goal kicks?

Confirm avoidance radius values for each field size or division

Until the ball is in play, all opponents must remain outside of this avoidance region, unless they are on their own goal line between the goalposts, and outside the penalty area for free kicks inside the opponents' penalty area.

All robots of the defensive team must immediately move away from the ball to outside the avoidance region when the free kick is awarded. Defensive robots that violate these restrictions are penalized with "Illegal Positioning" (see Section 13.5), which results in a standard removal penalty (see Section 13.2). Additional penalties against any further robots during the free kick, including pushing, do not result in an additional free kick but still incur the appropriate removal penalty.

14.3.3 Sanctions and Retakes

If, when a free kick is taken, an opponent is closer to the ball than the required distance, the opponent receives a removal penalty and the kick is retaken.

14.3.4 Completion

A free kick is deemed completed and play returns to normal when:

- The attacking team moves the ball clearly (except for a robot getting up, which is exempt from this rule), or
- The $30\,\mathrm{s}$ time period expires (or game time expires).

The head referee announces completion by calling “Ball Free”, and the GameController resumes the `playing` state.

14.3.5 Failed Free Kick

If the attacking team fails to execute the free kick within the allotted $30\,\mathrm{s}$, the free kick expires and play resumes normally. In this situation:

- The defending team may play the ball immediately after the free kick expires.
- If the defending team plays the ball first (before the originally attacking team), they may score a direct goal regardless of whether the original free kick was direct or indirect.

Example: The blue team is awarded a kick-in (indirect free kick) but does not play the ball within $30\,\mathrm{s}$. The referee calls “Ball Free”. A red robot immediately kicks the ball directly into the blue goal. The goal counts, as the failed-free-kick exception allows the defending team to score directly.

15 Kick-in / Throw-in

A kick-in (or throw-in) is an **indirect free kick** (see Section 14.2). A kick-in is awarded to the opponents of the player who last touched the ball when the whole of the ball leaves the field of play by crossing the touchline, either on the ground or in the air (see Section 2.1.8). Balls are deemed to be out based on the team that last touched the ball, irrespective of who actually kicked the ball.

15.0.1 Ball Placement

The ball is placed on the touchline at the point where it left the field.

15.0.2 Throw-in Option

Robots may also perform a throw-in with their hands instead of kicking. When performing a throw-in with hands, the robot must:

- Face the field of play.
- Have part of each foot either on the touchline or on the ground outside the touchline.
- Hold the ball with at least one hand.
- Deliver the ball from behind and over its head.
- Release the ball within 10 seconds.

If a robot attempts to perform a throw-in with hands and fails to respect these rules, a free kick is awarded to the opposing team.

15.0.3 Examples

Example 1: A blue robot at midfield kicks the ball over the left touchline $2\,\mathrm{m}$ into the red half of the field. The referee calls “Kick-in red” and the ball is replaced on the left touchline where it went out.

Example 2: A blue robot kicks the ball but the ball touches a red robot at midfield before leaving the field near the halfway line. The ball is regarded as out by red; the referee calls “Kick-in blue” and the ball is replaced on the touchline where it went out.

15.0.4 Procedural Requirements

All opponents must stand no less than $0.75\,\mathrm{m}$ (Small), $1.0\,\mathrm{m}$ (Medium), $1.5\,\mathrm{m}$ (Large) from the point at which the kick-in is taken. The ball is in play when it enters the field of play. After delivering the ball, the kicking robot must not touch the ball again until it has touched another player.

15.0.5 Offences and Sanctions

If, after the ball is in play, the kicker touches the ball again before it has touched another player:

- An indirect free kick is awarded to the opposing team, to be taken from the place where the infringement occurred.

If, after the ball is in play, the kicker deliberately handles the ball before it has touched another player:

- A direct free kick is awarded to the opposing team, to be taken from the place where the infringement occurred.
- A penalty kick is awarded if the infringement occurred inside the kicker's penalty area.

If an opponent unfairly distracts or impedes the kicker, they are cautioned for unsporting behaviour.

For any other infringement of this rule, the kick-in is taken by a player of the opposing team.

16 Goal Kick

A goal kick is a **direct free kick** (see Section 14.1) awarded when the whole of the ball passes over the goal line, either on the ground or in the air, having last touched a player of the attacking team, and a goal is not scored.

16.0.1 Ball Placement

The ball is placed on the corner of the goal area on the same side of the field that the ball went out. That is, the corner inside the field where the goal area line meets the goal line (not the T-junction on the goal line itself).

16.0.2 Avoidance Region

For a goal kick, the avoidance region is the entire penalty area. All robots of the opposing team must remain outside the penalty area until the ball is in play (see Section 14.3).

16.0.3 Example

Example: A blue robot kicks the ball out the end of the field to the right of the goal the red team is defending. The referee calls "Goal Kick red" and the ball is placed on the right corner of the goal area.

16.0.4 Procedural Requirements

The ball must be kicked directly out of the penalty area to be considered in play. If the ball is not kicked directly out of the penalty area from a goal kick, the kick is retaken.

16.0.5 Offences and Sanctions

If, after the ball is in play, the kicker touches the ball again before it has touched another player:

- An indirect free kick is awarded to the opposing team, to be taken from the place where the infringement occurred.

If, after the ball is in play, the kicker deliberately handles the ball before it has touched another player:

- A direct free kick is awarded to the opposing team, to be taken from the place where the infringement occurred.
- A penalty kick is awarded if the infringement occurred inside the kicker's penalty area.

In the event of any other infringement, the kick is retaken.

17 Corner Kick

A corner kick is a **direct free kick** (see Section 14.1) awarded when the whole of the ball passes over the goal line, either on the ground or in the air, having last touched a player of the defending team, and a goal is not scored.

17.0.1 Ball Placement

The ball is placed on the corner of the field on the same side that the ball went out.

17.0.2 Example

Example: The red goalkeeper kicks the ball out the end of the field to the right of the goal. The referee calls “Corner Kick blue”, the ball is placed on the corner to the right of the goal, and a free kick is started.

17.0.3 Offences and Sanctions

If, after the ball is in play, the kicker touches the ball again before it has touched another player:

- An indirect free kick is awarded to the opposing team, to be taken from the place where the infringement occurred.

If, after the ball is in play, the kicker deliberately handles the ball before it has touched another player:

- A direct free kick is awarded to the opposing team, to be taken from the place where the infringement occurred.
- A penalty kick is awarded if the infringement occurred inside the kicker’s penalty area.

In the event of any other infringement, the kick is retaken.

BELOW THIS LINE IS THE OLD STRUCTURE

WE MIGHT USE SOME OF IT IN THE FUTURE

====================================

17.1 Structure of the Game

17.2 Game Periods and Robot States

17.2.1 Game Periods

17.2.2 Robot States

17.3 Kick-off

17.3.1 Field-Side Selection and Initial Kick-off

17.3.2 Initial Kick-off

17.3.3 Ball in play

The ball is in play and kick-off ends once:

- it is touched by the attacking team and has visibly moved at least one ball radius from its initial position, or
- *KickOffBallFreeTime* seconds have elapsed in the `playing` state.

Move/define time constants

The GameController and head referee will indicate this by the call "Ball Free".

17.3.4 Ball out of play

The ball is out of play when:

- the ball has left the field of play, by wholly crossing the touchline or goal line (see Section 2.1.8), or
- play has been stopped by the referee.

The ball remains in play at all other times.

17.4 Goals

17.4.1 Goal Scored

A goal, including an own goal, is scored when the ball is considered *inside* the goal as defined in Section 2.1.8.[4]

The head referee signals a goal by a single whistle blow, followed by the call “Goal <color>”. The head referee should point with one arm towards the center of the field. To assist robots listening for whistles, the referee should blow the whistle from on the carpet at the end of the fields where the goal was scored.

After a team scores a goal, the game proceeds with a kick-off (see Section 17.3.2) for their opponents. The GameController signal (to the robots) of a goal being scored, will be delayed by $15\,\mathrm{s}$.

17.4.2 Invalid Goal

17.4.3 Competition Rules

17.5 Indirect Kick

17.5.1 Fallback mode

17.6 Penalty Kick

17.7 Game Stuck

17.7.1 Local Game Stuck

17.7.2 Global Game Stuck

17.8 Request for Pick-up

17.9 Timeout

17.9.1 Request for Timeout

Each team can call a **maximum of 1 timeout per game** with a total time of no more than **5 minutes**. During this time, both teams may change robots, change programs, or anything else that can be

[4]The goal line is part of the field.

done within the time allotted. During normal game time, a team may call a timeout at any stoppage of play (after a goal, stuck game, before a half, etc.). Alternatively, a team may call a timeout before a penalty shootout if they have not used their timeout yet (see **??**).

The timeout ends when the team that called the timeout says they are finished, at which time they must be ready to play. The other team must be ready to play at the time the timeout runs out, or **2 minutes** after a prematurely called end of the timeout, whichever is earlier. If the other team is not ready to play in time, it has to call a timeout of its own.

The clock stops during timeouts, even during the preliminaries, and is reset to the time when the current stoppage of play began.

After the completion of the timeout, the game resumes with a kick-off for the team which did not call the timeout.

If a team is not ready to play at the assigned time for a game, the referee will call the timeout for that team. After the expiration of such a timeout, if the team is still not ready to play then the referee shall start the game with only one team on the field. The team that was not ready can return its robots to the field as per the rules for “Request for Pick-up”. If both teams are not ready, the referee will call timeouts for both teams. This “double timeout” expires after 10 minutes.

17.9.2 Referee Timeout

17.10 Extra Time

17.11 Mercy Rule

A game will conclude once the game score shows a goal difference of 10. Ending the game is mandatory once a goal difference of 10 is reached.

17.12 Drop Ball Rule

17.13 Ball Stop Rule

17.14 Determine the Winner of a Match

17.14.1 Winning Team

17.14.2 Winner after Drawn

17.15 Penalty Kick Shoot-Out

17.15.1 Penalty Kick

17.15.2 Sudden Death Shoot-Out